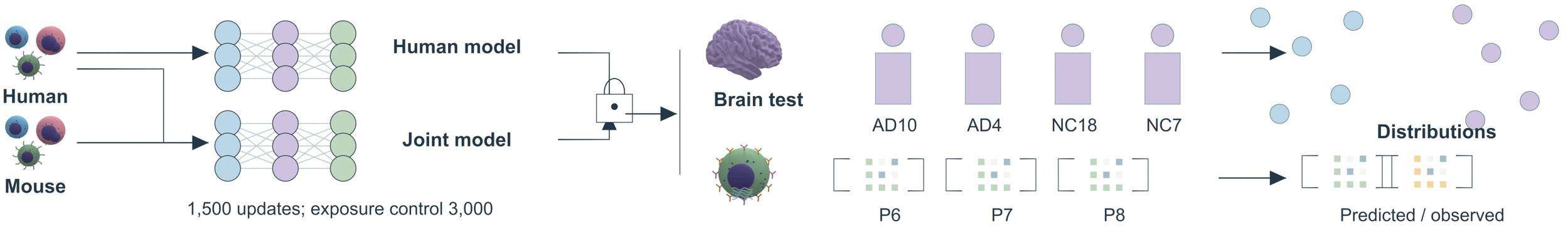
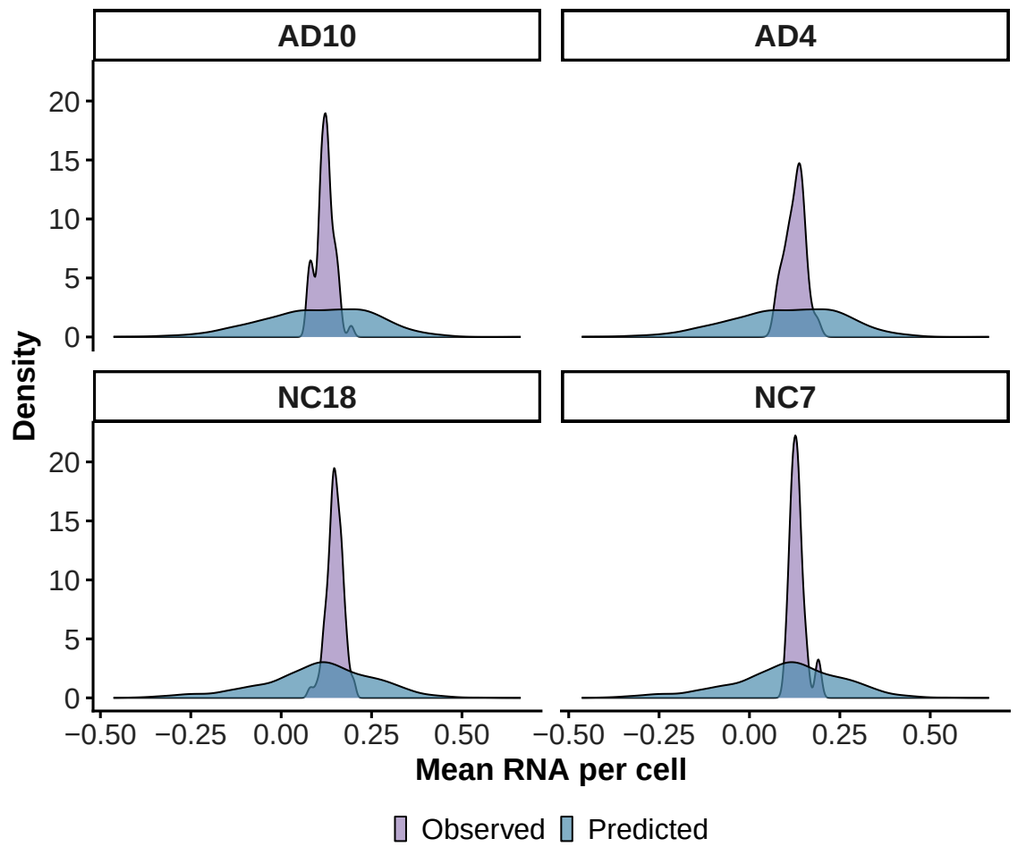
A



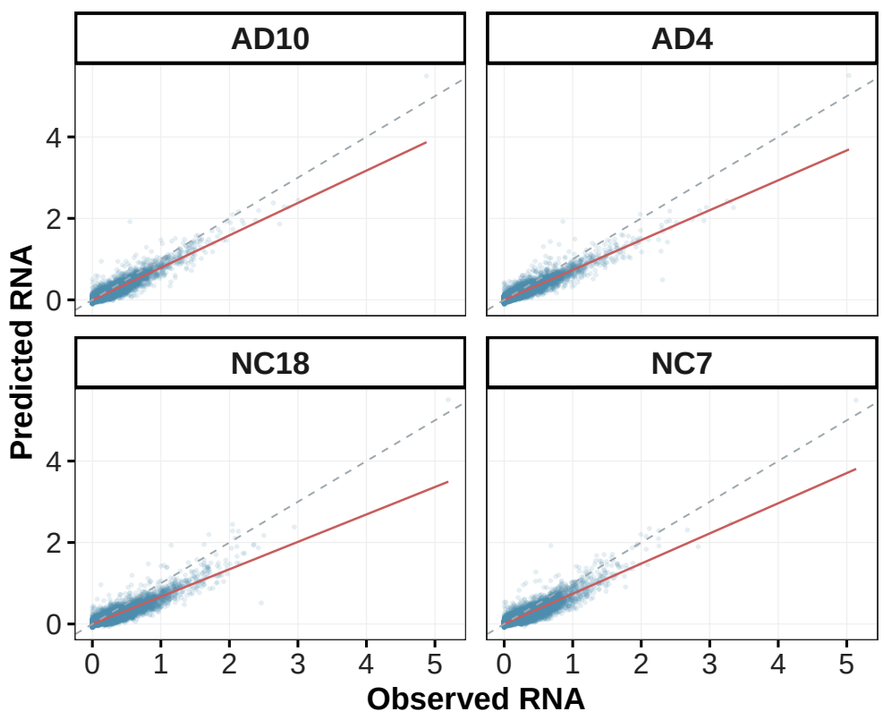
B

	Brain MMD	Energy distance	RNA MSE	RNA r	ADT MSE	ADT r
Joint model	0.129	0.184	0.051	0.739	0.413	0.410
Human model	0.126	0.189	0.052	0.719	0.413	0.412
Human: projection	0.129	0.191	0.057	0.685	0.401	0.448
Human: no protein	0.121	0.185	0.047	0.755	NA	NA
Deterministic	0.178	0.300	0.079	0.664	0.426	0.389
Detached protein head	0.128	0.185	0.052	0.719	0.579	0.246
Exposure matched (3,000)	0.119	0.180	0.051	0.722	0.408	0.436
Learned projection	0.121	0.177	0.052	0.718	0.395	0.447
Random−pair objective	0.322	0.548	0.046	0.764	0.410	0.426
MLP dynamics	0.133	0.190	0.053	0.725	0.410	0.413
Swapped mouse labels	0.132	0.187	0.051	0.730	0.414	0.409
No blood expression	0.119	0.177	0.052	0.752	0.407	0.423
No disease condition	0.131	0.186	0.051	0.739	0.411	0.417
No protein head	0.127	0.186	0.053	0.703	NA	NA
No reconstruction	0.119	0.191	0.057	0.701	0.416	0.407
No species condition	0.132	0.188	0.052	0.735	0.413	0.414
Shuffled protein labels	0.130	0.187	0.052	0.730	0.417	0.401
Protein weight 0.1	0.130	0.191	0.055	0.706	0.443	0.368
Random projection	0.128	0.181	0.050	0.735	0.409	0.426
Shuffled ESM	0.129	0.183	0.050	0.746	0.410	0.411

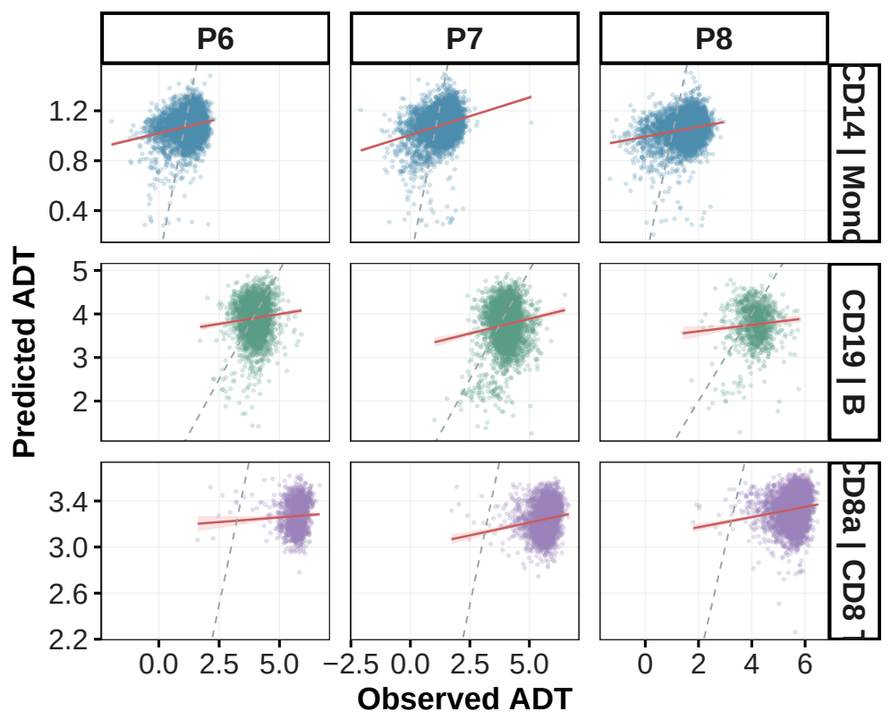
C



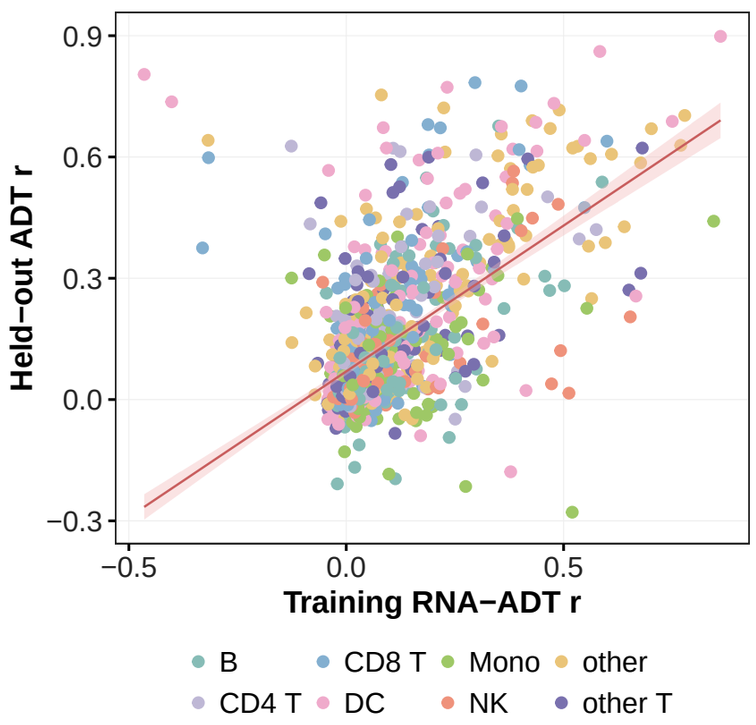
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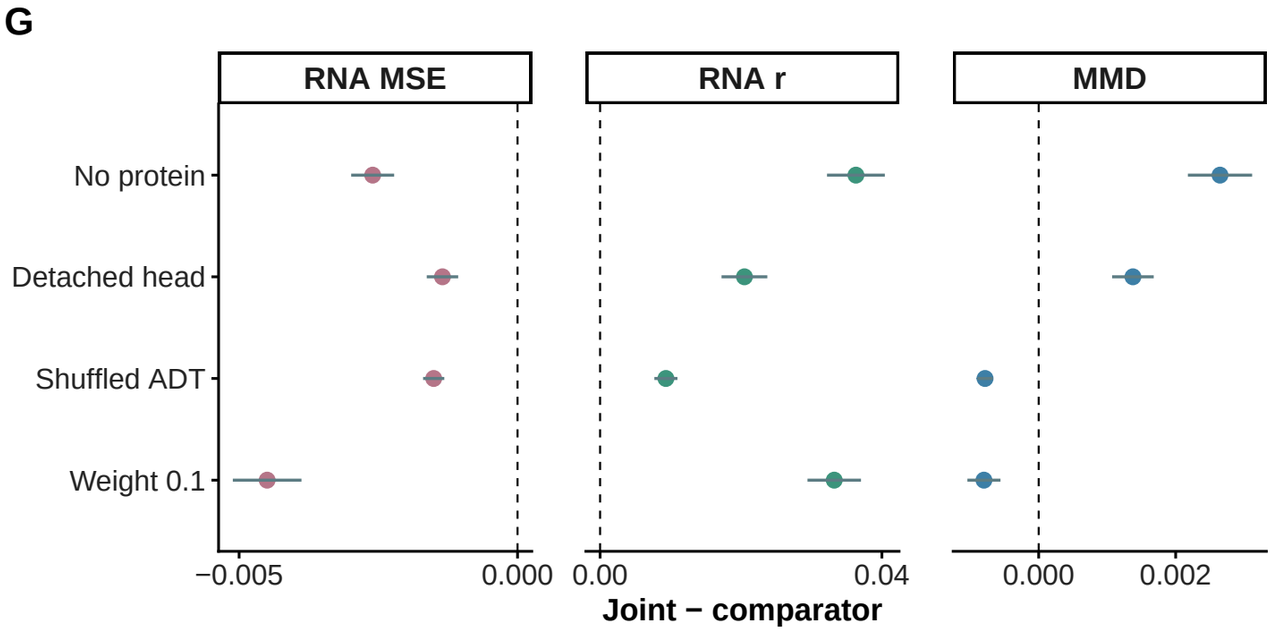
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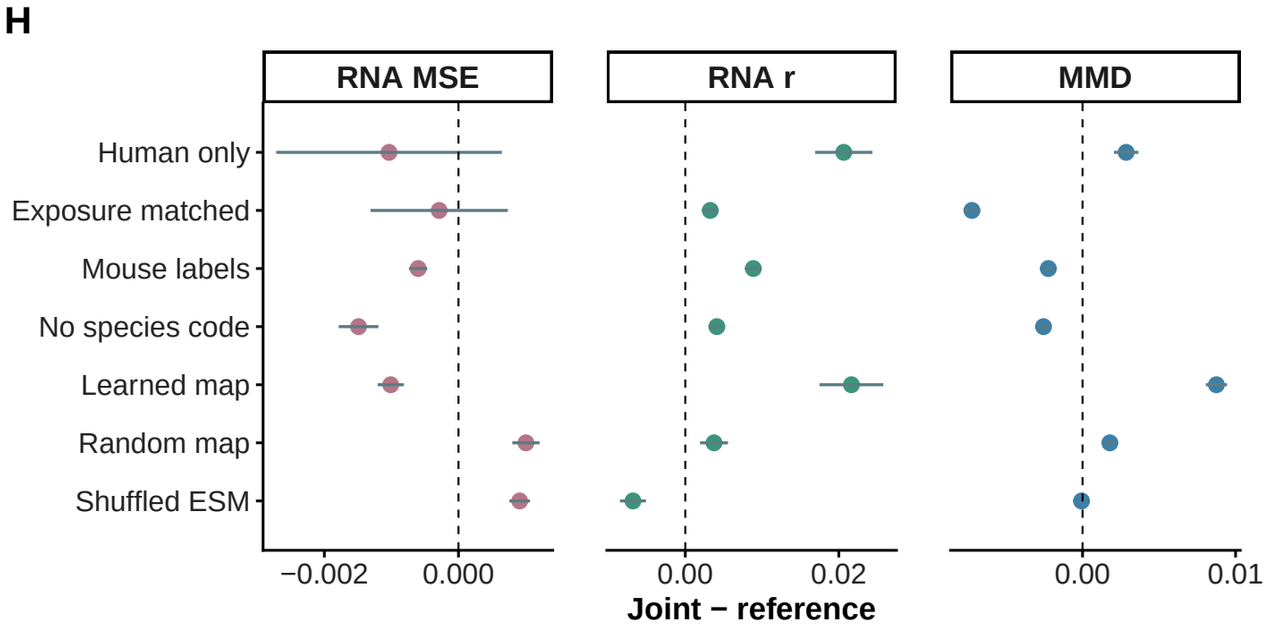
F



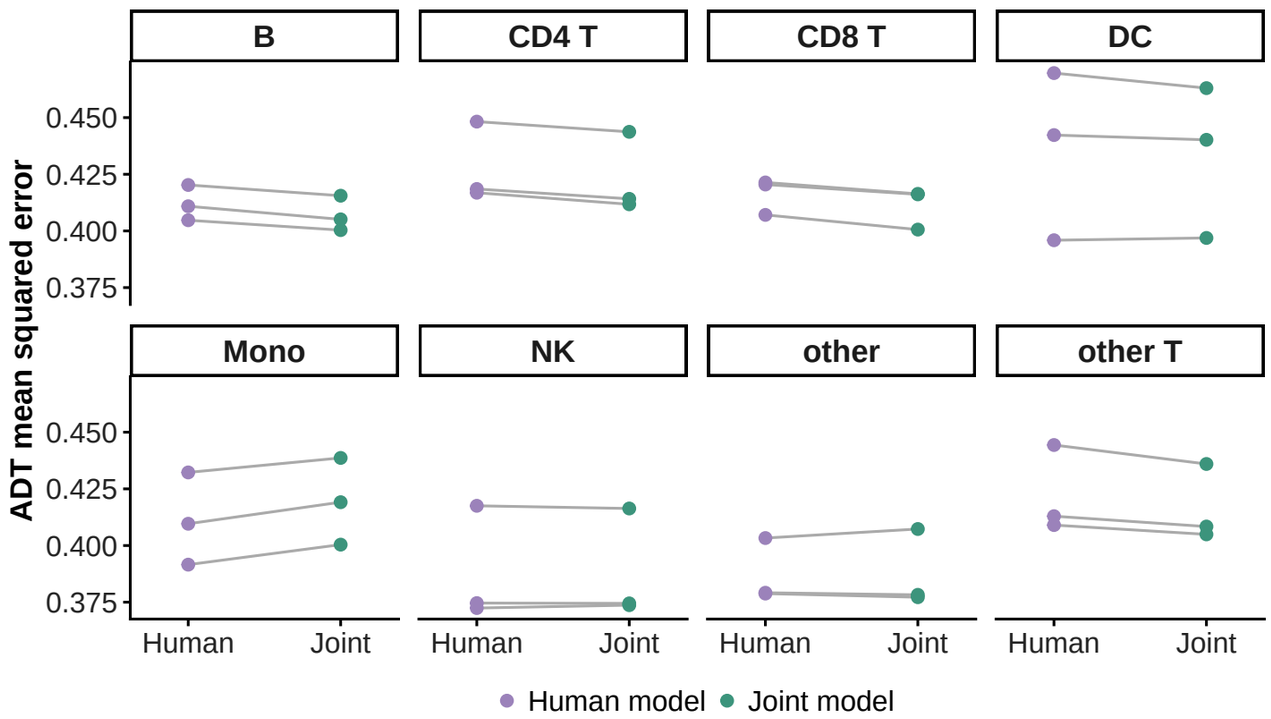
G



H



I



J

